Assignment 1: Create Your Own Game in Unity

Updated by 2026/1/26 15:15. Updated content is highlighted with blue marks.

Overview

After three hands-on lab sessions, you should now feel comfortable with the basics of Unity and confident in building interactive experiences. In Lab 1 and Lab 2, you learned how to create a complete 3D game by working with scenes, materials, lighting, physics, animations, C# scripts, UI elements, and audio. In Lab 3, you took things a step further by developing an augmented reality (AR) game using Vuforia Engine, exploring how digital content can interact with the real world.

These labs were designed to help you build both technical confidence and an initial understanding of interaction design within Unity. This assignment continues that learning journey by encouraging you to reflect on, refine, and build upon what you have learned so far. So, let's get started !

Assignment Tasks

For this assignment, **you can select one of the following two independent tasks**. You may choose the option that best matches your interests and the work you have completed during the labs:

Option 1: Simple 3D Game (Lab 1 + Lab 2)

For task 1, submit a game based on the **Lab 1 and Lab** 2 tutorial content. You need to first follow the **Lab 1 and Lab 2** tutorial videos and complete the full 3D game as demonstrated in the tutorials. After completing the tutorial-based game, you may optionally enhance or extend the game by adding new features, improving interaction design, refining visuals or audio feedback, or introducing creative gameplay ideas. These improvements are not required, but will be considered for bonus credit.

Evaluation

Total points of assignment 1 are up to 10 points.

Total score = basic points (7 pts) + bonus points (3 pts).

Basic

1. Correctly add physical interactions such as gravity, rigid bodies, and colliders to the game (video 5) - 1 point
2. Creating basic animations and ran successfully (video 6) - 1 point
3. Implement first person player movement and camera control (video 8) - 1 point
4. Using canvas to implement text function of video 10 (video 9-11) - 1 point
5. Using canvas to implement button function of video 12 (video 9-12) - 1 point
6. Adding music to the game correctly (video 14) - 1 point
7. Build an executable file that runs correctly (video 16) - 1 point

Bonus

Subject to meeting the basic requirements, you are encouraged to expand upon and refine the game in any way you see fit. Whether through **UI design, additional levels,**

improved controls, sound design, or any other enhancements that elevate the overall experience—such efforts will be considered for bonus points. Notice that any improvements that are neither written down in the **readme.txt** nor discovered will not be counted as bonuses.

Bonus points will be awarded based on the scope and quality of the improvements made, with the bonus score **not exceeding 3 points**.

Important Notes on Evaluation

Please note that grading will be primarily based on the **executable build (.exe) and the README file**. The Unity project file will be used for reference only and may not be opened during grading.

Therefore, it is your responsibility to ensure that the executable file functions correctly and fully demonstrates all the features and interactions you have implemented. Any features not shown in the executable file may not be included in the score.

Submission Requirements

You are required to submit the following 3 items to the Assignment 1 portal on Blackboard:

- Unity project folder (.zip)
- Unity executable build folder (.zip)
- README document (.txt)

Please do not combine these files into a single compressed package. Each file should be uploaded individually.

- **Unity project folder (.zip)**
  - Compress your full Unity project folder into a .zip file
  - This should include all assets, scenes, scripts, Animator Controller, etc.
  - File name format: YourName&ID_UnityProject.zip

- **Unity executable build folder (.zip)**

  After building your project in Unity, you will get a folder that contains your .exe file. You can refer to the appendix for the complete build process.

  - Compress this entire folder into a .zip file
  - File name format: YourName&ID_Build.zip
  - This is the version we will run and test

- **README document (.txt)**

  Include a plain text file named README.txt that clearly states:

  - Which assignment task you selected (**Task 1: 3D Simple Game** or **Task 2: AR Game**)
  - The operating system used to develop and run the project (**Windows or macOS**). If you are using macOS, please indicate that the game is built for macOS in your readme.txt file. Maybe like: “This game was built using macOS. Please run the .app file to play the game on macOS.”
  - If you make any **improvements in your game for bonus points**, please

briefly introduce your improvement in your readme.txt file. For example: “I added 2 more levels after the example level.”

Option 2: AR Game (Lab 3)

You should first follow the **Lab 3** tutorial videos and complete the full AR game using Vuforia Engine as demonstrated in the lab. The basic AR setup, tracking, and interactions shown in the tutorial should be fully implemented. After completing the tutorial-based AR game, you may optionally optimize or enhance your project, such as improving AR interactions, refining visual presentation, or adding creative AR elements. These enhancements are optional and will be evaluated for bonus credit.

Evaluation

Total points of assignment 1 are up to 10 points.

Total score = basic points (7 pts) + bonus points (3 pts).

Basic

1. **Set up Vuforia Multiple Image Target.** - (video1&2) 1point

   Recognize two images and show models of Astronaut and Drone.
2. **Clicked AR Button** - (video2) 1point

   Click Button on Drone and it shoots projectile from spawn point.
3. **Projectile Collider detection** - (video2) 1point

   When a projectile shoots at an Astronaut, destroy the Astronaut model.
4. **Correct set Astronaut and Drone card collision and trigger Astronaut animation.** - (video3) 2point

   When Astronaut and Drone Card collide, Astronaut should do Drill animation.
5. **Correct set Water and Lava card collision and trigger Rock model.** - (video4) 1point

   When Water and Lava cards collide, show only Rock on the screen.
6. **Add Fusion effect for generating Rock.** - (video4) 1point

   When water and lava cards collide, show the fusion effect first before rock is displayed.

Bonus

Subject to meeting the basic requirements, you are encouraged to expand upon and refine the game in any way you see fit. Whether through **UI design, additional levels, improved controls, sound design, or any other enhancements that elevate the overall experience**—such efforts will be considered for bonus points.

Notice that any improvements that are neither written down in the **readme.txt** nor discovered will not be counted as bonuses.Bonus points will be awarded based on the scope and quality of the improvements made, with the bonus score **not exceeding 3 points**.

Hits: You can try to merge three games into one like when an astronaut is close to a rock, it will do drill animation. Or you can try that when a drone is close to an astronaut, the drone will shoot automatically.

Important Notes on Evaluation

Please note that grading will be primarily based on the **gameplay video and the README file**. The Unity project file will be used for reference only and may not be opened during grading.

Therefore, it is your responsibility to ensure that the video clearly and completely demonstrates all the features and interactions you have implemented. Any functionality not shown in the video may not be considered in the assessment.

Submission Requirements

You are required to submit the following 3 items to the Assignment 1 portal on Blackboard:

- Unity project folder (.zip)
- Gameplay Screen Recording Video (.mp4)
- README document (.txt).

Please do not combine these files into a single compressed package. Each file should be uploaded individually.

- **Unity project folder (.zip)**
  - Compress your full Unity project folder into a .zip file
  - This should include all assets, scenes, scripts, Animator Controller, etc.
  - File name format: YourName&ID_UnityProject.zip

- **Gameplay Screen Recording Video (.mp4)**

  Submit a screen recording video that demonstrates your completed game.

  - For AR projects, the video must show the actual AR interaction and clearly demonstrate all implemented features.
  - The video should be continuous and complete, without missing key interactions or gameplay elements.
  - File name format: YourName&ID_ARGame.mp4

- **README document (.txt)**

  Include a plain text file named README.txt that clearly states:

  - Which assignment task you selected (**Task 1: 3D Simple Game** or **Task 2: AR Game**)
  - The operating system used to develop and run the project (**Windows or macOS**). If you are using macOS, please indicate that the game is built for macOS in your readme.txt file. Maybe like: “This game was built using macOS. Please run the .app file to play the game on macOS.”
  - If you make any **improvements in your game for bonus points**, please briefly introduce your improvement in your readme.txt file. For example: “I added 2 more levels after the example level.”

Due Date

- **23:59, Feb. 1st (Sunday)**
- **Late submissions will result in a point deduction.**

Late submissions will result in point deductions: 1 point per day of delay (e.g., 1 point for one day late, 2 points for two days late, and so on.

Appendix

- Open your Unity project.
- Before building the executable file, save your project first. Click **File>Save** and **File>Save Project**
- Go to **File > Build Settings**.
- In the **Platform** section, select the appropriate platform based on your operating system:
  - **Windows users**: Select PC, Mac & Linux and ensure that **Target Platform** is set to Windows.
  - **Mac users**: Select PC, Mac & Linux and ensure that **Target Platform** is set to macOS

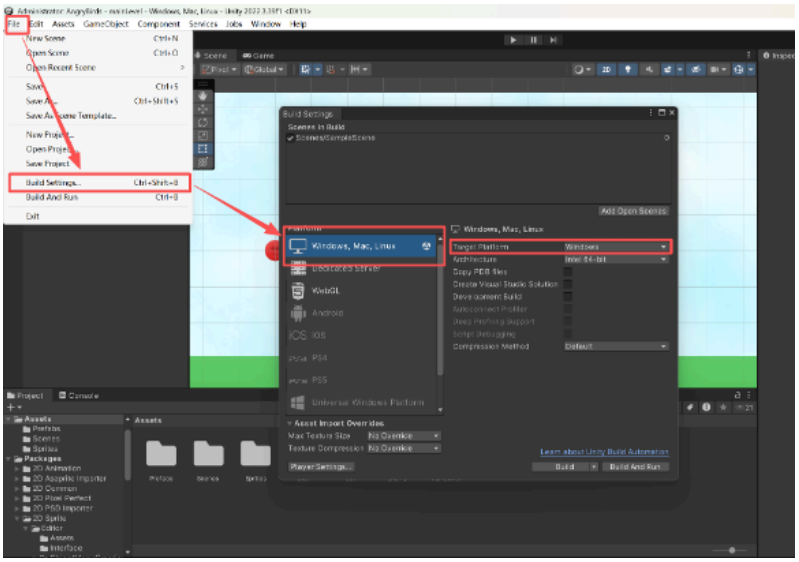

Fig. 1 Screen shot of Build settings

- Ensure your game scene is added in the **Scenes In Build** section (click Add **Open Scenes** if necessary).

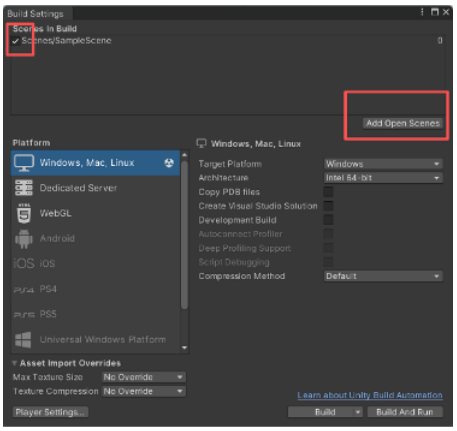

Fig. 2 Screen shot of Scenes In Build section

- Give your product a name. You can change your game's name in **Build Settings -> Player Settings -> Product Name**. This name will be the file name of your final executable file (e.g., "AngryBirdsReplica.exe" for Windows or "AngryBirdsReplica.app" for macOS).

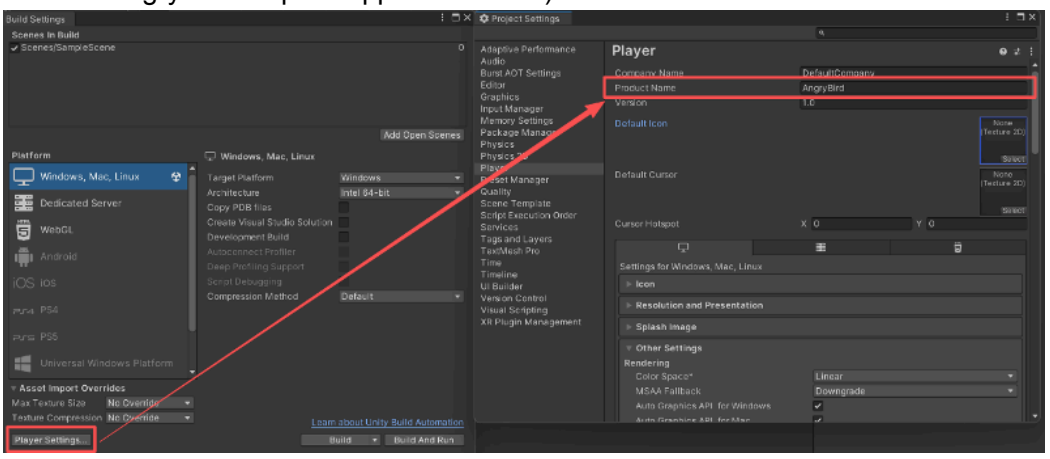

Fig. 3 Screen shot of Product Name setting

- Click **Build**, choose a location and create **a build folder named "run"** to save the build.

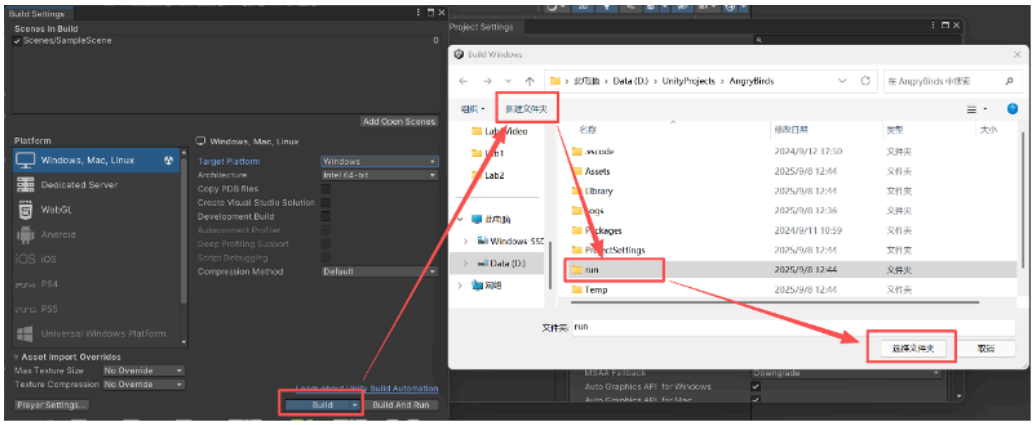

Fig. 4 Screen shot of build

- Unity will create a folder containing executable and necessary data files.